



OrBit User Guide

Module 04: 3-Way Conflict Detection & Resolution

PUBLIC DOCUMENTATION — FOR DEVELOPERS USERS

1 Understanding Conflicts in Local-First Environments

In a decentralized peer-to-peer developer mesh, two developers may modify the exact same lines of a file concurrently while disconnected or before a P2P push reaches them.

OrBit resolves concurrent modifications using a formal **3-Way Merge Algorithm** implemented in `crates/sync-diff/src/conflict.rs`.

2 The 3-Way Conflict Algorithm

When an incoming remote delta arrives from a peer:

1. **Base Snapshot (O):** The common ancestor baseline snapshot on the `main` branch.
2. **Local Workspace (A):** The current state of your local working directory on the `sub` branch.
3. **Remote Payload (B):** The incoming delta pushed by your teammate.

The `sync-diff` conflict engine checks for overlapping file modifications:

- If File X modified in B was untouched in A , changes are applied cleanly automatically.
- If File X was modified in both A and B at non-overlapping line ranges, OrBit auto-merges the changes.
- If File X was modified in both A and B at overlapping line ranges, a **Conflict Event** is declared.

3 The Visual Conflict Resolver Modal

When a conflict occurs, OrBit prevents file corruption by pausing auto-apply and launching the **Conflict Resolver Modal** in the OrBit Desktop Client.

Visual Conflict Inspector GUI	
Local Changes (Your Code)	Incoming Peer Changes (Teammate)
const port = 8080; const host = "localhost";	const port = 9090; const host = "0.0.0.0";

Figure 1: Side-by-Side Conflict Comparison UI

4 Conflict Resolution Strategies

Inside the Conflict Resolver Modal, developers can choose between three distinct resolution strategies:

- 1. **Accept Local ("Keep Mine")**: Discards the incoming remote changes for the conflicting file regions and preserves your local code.
- 2. **Accept Remote ("Keep Teammate's")**: Overwrites conflicting local regions with the incoming peer code.
- 3. **Manual Merge / Conflict Markers**: Inserts standard Git conflict block markers directly into the file:

```
1 <<<<<<< LOCAL (Your Changes)
2 const port = 8080;
3 =====
4 const port = 9090;
5 >>>>>>> REMOTE (Peer ed25519:7f8a3b...)
```

You can then edit the file directly in VS Code, save it, and click **Finalize Merge** in the Desktop UI.

Pre-emptive Conflict Warnings

To avoid conflicts altogether, install the OrBit VS Code extension. It warns you in real-time if a teammate is currently editing the exact same file in their IDE!